When to Use

To compare the means of two or more quantitative variables obtained from dependent samples (repeated measures or matched groups). The two or more scores might be the same variable measured at different times or under different conditions, comparable variables measured at the same time, or some combination.

Research Hypothesis: There are significant differences in the three types of engagement scores (emotional impact, aesthetic appeal, cognitive engagement) for Superman media portrayals, with emotional impact receiving the highest scores and cognitive engagement receiving the lowest.

Null Hypothesis: Superman media portrayals have the same mean scores for emotional impact, aesthetic appeal, and cognitive engagement.

Research Design

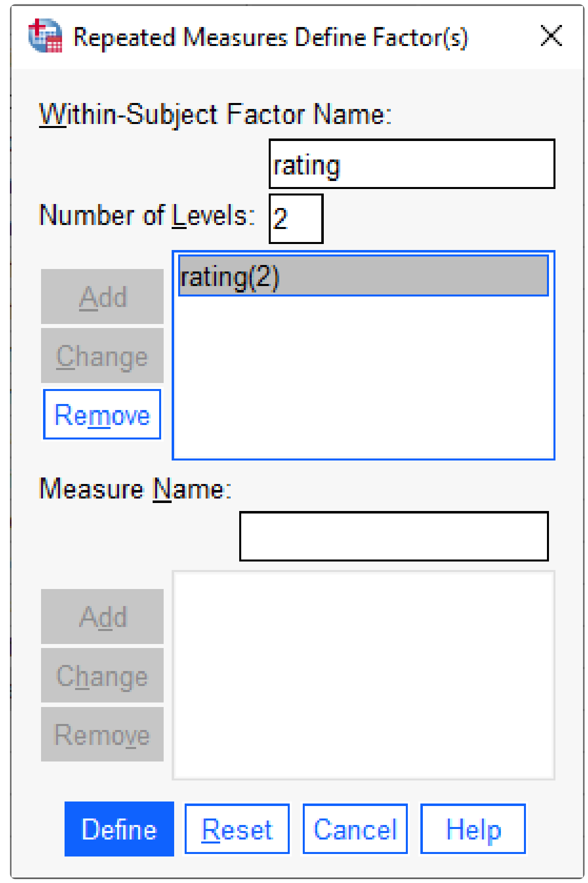
The IV is Engagement Type, with the conditions Emotional Impact, Aesthetic Appeal & Cognitive Engagement. The DV is the score on each engagement measure.

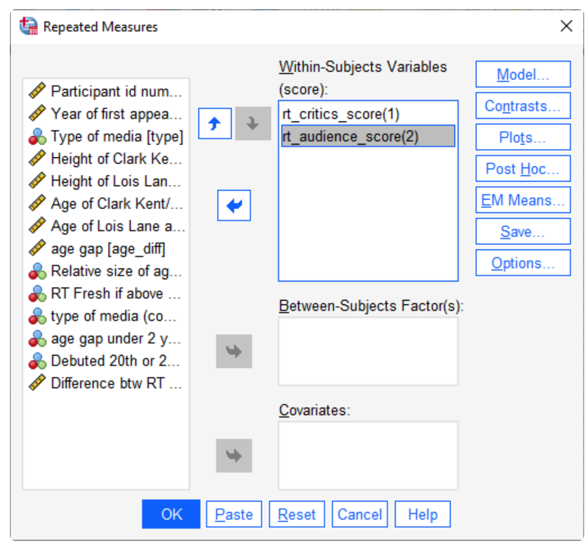
Emotional Impact	Aesthetic Appeal	Cognitive Engagement

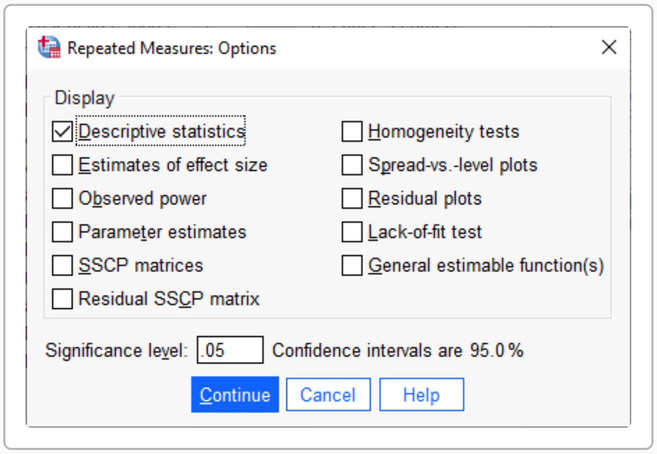
Variables in the Analysis: In a WG design the variables in the analysis are the variables holding the DV scores for each IV condition (emotional_impact, aesthetic_appeal & cognitive_engagement).

Analyze → General Linear Model → Repeated Measures

- In the **Repeated Measures Definition Window** enter your name for the IV in the “Within-subject Factor Name” box (EngagementType)
- Enter the number of conditions of the IV in the “Number of levels” window (3)
- Click the “Add” button
- Click the “Define” button
- In the **Repeated Measures** window highlight the variables that are the DV score for each condition and click the arrow
- Click the “Options” button — in the **Repeated Measures: Options** window check the “Descriptives” box







SPSS Syntax

```
GLM emotional_impact aesthetic_appeal cognitive_engagement   ①
  /WSFACTOR=EngagementType 3                                 ②
  /PRINT=DESCRIPTIVE.                                        ③
```

① List variables holding DV for each IV condition
② Give a name to the WG IV & tell # conditions
③ Get mean and std

SPSS Output

Descriptive Statistics

	Mean	Std. Deviation	N
emotional_impact	13.27	3.385	44
aesthetic_appeal	9.80	2.833	44
cognitive_engagement	4.36	1.170	44

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F
Sig.					
EngagementType	Sphericity Assumed	1776.110	2	888.055	138.518
<.001	Error(EngagementType)	Sphericity Assumed	551.357	86	6.411

SPSS provides different "versions" of the ANOVA output. We will focus on the "traditional" analysis, which SPSS labels as "Sphericity Assumed".

- **df effect** & **F** & **p-value** to use
- **df(error)** & **Mean Square Error** term for this analysis

There is a significant difference among the means of the three different types of engagement scores, $F(2, 86) = 138.518$, $p < .001$.

Steps for computing and interpreting LSD minimum mean difference

1. Determine df(error) for the analysis: **df(error) = 86**
2. Use the t-table to determine the critical value of t (at p = .05) for this df(error): **t-critical = 1.99**
3. Determine the MSerror for the analysis: **MSerror = 6.41**
4. Determine the number of participants in the analysis: **n = 44**
5. Apply the LSD formula to obtain the minimum mean difference

$$d_{LSD} = \frac{t \times \sqrt{2 \times MSE_{error}}}{\sqrt{n}} = \frac{1.99 \times \sqrt{2 \times 6.41}}{\sqrt{44}} = 1.07$$

6. Compare the minimum mean difference (1.07) with each pairwise mean difference:

- If the pairwise mean difference is larger than the minimum mean difference, then those two conditions have means that are “significantly different”
- If the pairwise mean difference is smaller than the minimum mean difference, then those two conditions have means that are “statistically equivalent”

For this analysis — compare the minimum mean difference of 1.07 with each of the pairwise differences:

Pair	Mean Difference	LSD Result
Emotional Impact vs Aesthetic Appeal	3.48	Significant
Emotional Impact vs Cognitive Engagement	8.91	Significant
Aesthetic Appeal vs Cognitive Engagement	5.44	Significant

Reporting the Results

Sample APA Write-Up

The engagement scores for Superman media portrayals are summarized in Table 1. Emotional Impact (*M* = 13.27, *SD* = 3.39), Aesthetic Appeal (*M* = 9.80, *SD* = 2.83), Cognitive Engagement (*M* = 4.36, *SD* = 1.17). There was a significant difference among the three engagement score distributions, $F(2,86) = 138.52$, $MSe = 6.41$, $p < .001$. Pairwise comparisons using LSD (with a minimum mean difference = 1.07) revealed that Emotional Impact was rated higher than Aesthetic Appeal, and Emotional Impact was rated higher than Cognitive Engagement, and Aesthetic Appeal was rated higher than Cognitive Engagement.

Table 1.

Summary of engagement scores for Superman media portrayals.

	Mean	SD
Emotional Impact	13.27	3.39
Aesthetic Appeal	9.80	2.83
Cognitive Engagement	4.36	1.17